# Phase 1 (O1): baseline detection on MQTTset

Evaluating and Improving the Adversarial Robustness of ML-Based Intrusion Detection in MQTT IoT Networks

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## 1 · WHAT THE DATA SHOWED

- **Heavy duplication.** 63.31% of rows are exact duplicates. Normal for packet-level traffic, but in the **published train/test split, 69.59% of test rows have an exact copy in training** (98.43% for `legitimate`).
- **Label ambiguity caps accuracy.** 1,335 identical feature vectors carry more than one label. **99.89%** of those rows are bare TCP packets with no MQTT layer, labelled by capture file. Unavoidable error is 6.73%, so **no model can exceed 93.27% accuracy** on the 20 features.
- **`mqtt.msg` dropped.** Payload text (50,054 values) is memorisable and attacker-controlled, and every `slowite` row, and no other row, writes it as `0.0`: a formatting giveaway.
- **Features used: 20** = 33 − 12 constant − `mqtt.msg`. Hex fields → integers; "field absent" kept distinct from zero.

## 2 · RESULTS: HELD-OUT TEST SET, MACRO F1 (HEADLINE)

| Model | A: published | B: de-dup. | B: CV, 15 folds |
|---|---|---|---|
| **XGBoost** | **0.6164** | **0.8673** | **0.8662 ± 0.0174** |
| Random Forest | 0.6094 | 0.8559 | 0.8553 ± 0.0178 |
| MLP | 0.5837 | 0.8030 | 0.8005 ± 0.0247 |

Balanced class weights, no oversampling. B = one row per (feature vector, label), grouped + stratified 70/30 split, zero vectors shared between train and test.

| Cost (Mac, Variant B) | Size | ms / packet |
|---|---|---|
| **XGBoost** | **1,080 KB** | **0.106** |
| Random Forest | 42,049 KB | 2.985 |
| MLP | 123 KB | 0.164 |

Latency = one packet at a time, single thread. XGBoost is the best model on accuracy and cost.

## 3 · MAIN FINDING: THE LEAKY SPLIT SCORES *LOWER*, AND WHY

Variant A's test set is three populations (Random Forest, balanced weights):

| seen, one label · 34.37% · acc 99.98% | seen, ambiguous · 35.21% · acc 57.54% | unseen · 30.41% · acc 98.46% |
|---|---|---|

**Memorisation is real:** a third of the published test set is answered almost perfectly because it was in training. But an even larger block of ambiguous TCP packets pulls Variant A down, so **the A − B gap is not a measure of inflation**. The three-group breakdown is the honest measure. Without class weights, A reaches 92.14–92.34% accuracy, close to the 93.27% ceiling.

## 4 · DECISIONS I NEED YOUR VIEW ON

1. **Framing:** report the three-group breakdown as the headline, instead of the A − B gap.
2. **Class weights:** keep balanced weights as the main setting; report no-weights as a sensitivity check (fewer false alarms, lower rare-class recall).
3. **Drop `mqtt.msg`** (removes it from the O2 constraint set).
4. **`flood`** has only 80 unique test vectors in B (94 in A): report its per-class scores as unreliable.

## 5 · NEXT (PHASE 2, O2)

- Strongest XGBoost feature is `mqtt.msgid`. It is non-zero only when QoS > 0, which in this data is almost only `dos`. **An attacker can choose QoS 0**: a first constraint-aware evasion to test.
- Classify all 20 features as mutable / immutable / dependent; set up a Mosquitto broker to validate perturbations.
- Before comparing with published 97–99%: check which MQTTset file and features each paper used.

All numbers from code run on my MacBook (5 Nov 2026; 10.5 min; seed 42; Python 3.14.2, scikit-learn 1.9.1, xgboost 3.4.1). A cloud re-run on identical data gave identical Random Forest and XGBoost results; only the unstable MLP on Variant A differed slightly between machines. Code, logs and full tables: GitHub `rasheedali78624/finalyearproject`, branch `claude/amazing-hopper-glemny`, `docs/phase1/PHASE1_FINDINGS.md`.